

have longer corresponding focal lengths. So when you zoom in, you increase the focal length and when you zoom out, you decrease it.

The focal length (f), the distance from the front nodal point to the object to be photographed (S_1), and the distance from the rear nodal point to the image plane (S_2) are then related by:

$$\frac{1}{S_1} + \frac{1}{S_2} = \frac{1}{f}$$

As S_1 is decreased, S_2 must be increased. For example, consider a normal lens for a 35 mm camera with a focal length of $f = 50$ mm. To focus a distant object (), the rear nodal point of the lens must be located a distance $S_2 = 50$ mm from the image plane. To focus an object 1 m away ($S_1 = 1000$ mm), the lens must be moved 2.6 mm further away from the image plane, to $S_2 = 52.6$ mm.

As you would have gauged by now, apart from determining the view field, focal length also relates to the magnification of the image. A lens with a focal length about equal to the diagonal size of the film (about 35 mm) or sensor format is known as a normal lens. Normal lens ensures that the view field matches the typical view field of the human eye. As in, the objects perceived via the view field would match how a human eye would see the scene. For full-frame 35mm-format cameras, the diagonal is 43 mm and a typical “normal” lens has a 50mm focal length. A lens with a focal length shorter than normal is often referred to as a wide-angle lens (typically 35 mm and less, for 35mm-format cameras), while a lens significantly longer than normal may be referred to as a telephoto lens (typically 85 mm and more, for 35mm format cameras).



Courtesy: Nikon

The difference in view at different focal lengths

To be very strict about the terminology use, the long focal length lenses are truly “telephoto” only if the focal length is longer than the physical length of the lens, but the term is often used to describe any long focal length lens.

Due to the popularity of the 35mm standard, camera–lens combinations are often described in terms of their 35mm equivalent focal length, that is, the focal length of a lens that would have the same angle of view,